

Ismail AISSAOUI

State Engineer in Artificial Intelligence (ESI-SBA)

Email: ismail.aissaoui@example.com

Phone: [Your Phone Number]

LinkedIn: [Your Link]

Professor Stephan Günnemann
Technical University of Munich (TUM)
Data Analytics and Machine Learning Group
Munich, Germany

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Subject: Application for Doctoral Research - Robust Sequential Graph ML

Dear Professor Günnemann,

I am writing to formally submit my application for a doctoral position within the Data Analytics and Machine Learning group at the Technical University of Munich. As an AI Engineer specializing in advanced neural architectures for complex systems, I have long admired your work on robust graph learning and uncertainty estimation, and I am eager to contribute to your group's research.

In my current role at **Sonatrach**, I am leading the development of a **Generative Digital Twin** for gas turbine health monitoring. This project has required me to move beyond standard deep learning paradigms to ensure model robustness in high-stakes industrial environments. I have been specifically investigating **State Space Models (Mamba-SSM)** and **xLSTM** as scalable alternatives to Transformers, finding that their ability to handle long-range dependencies with linear complexity is critical for real-time industrial sequences.

I am particularly interested in your research on the robustness of **Graph Neural Networks (GNNs)**. My background includes applying GNNs to model structural dependencies and sensor networks, and I am keen to explore how these graph-based models can be combined with SSMs to create "Robust Spatial-Temporal" architectures. Furthermore, my experience with **Physics-Informed Neural Networks (PINNs)** has given me a unique perspective on how to use domain-specific constraints to regularize latent representations, a topic that I believe aligns with your interest in theoretically sound machine learning.

TUM is my top choice for a PhD due to its reputation as a leader in computation and information technology. I am a self-driven researcher with a strong background in Python and PyTorch, and I am ready to commit to the rigorous academic standards of your laboratory.

I have attached my academic transcripts, my CV (from H+ recognized ESI-SBA), and a research proposal for your review. I would be honored to discuss how my technical skills and passion for robust ML can contribute to your group.

Thank you for your time and consideration.

Sincerely,

Ismail AISSAOUI